Intraoperative functional and metabolic brain mapping using hyperspectral imaging

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ABSTRACT

The acquisition of narrow wavelength bands in the Near-Infrared (NIR) range using hyperspectral imaging have the potential to yield functional and metabolic information during a neurosurgical operation. The analysis of the reflectance spectra through the modified Beer-Lambert law and Monte Carlo simulations enable us to measure the concentration changes of oxy, deoxygenated hemoglobin and cytochrome-c-oxidase during the video acquisition. A functional model has been implemented to evaluate the functional brain areas following neuronal activation by physiological stimuli. The results show a good correlation between the computed quantitative functional maps and the brain areas identified by electrical brain stimulation. This work demonstrates that a quantitative modeling of the brain hemodynamic and metabolic biomarkers could evaluate in a robust way the functional areas and the cellular energy metabolism during neurosurgery.

Keywords: functional brain mapping, metabolic brain mapping, Monte-Carlo simulations, intraoperative imaging, hyperspectral camera

1. INTRODUCTION

Chance et al. have demonstrated that optical imaging can be used to monitor brain activity with the determination of concentration changes of oxygenated hemoglobin ΔC_{HbO_2}, deoxygenated hemoglobin ΔC_{Hb} and oxygenated cytochrome-c-oxidase ΔC_{oxCCO}.[1] Since then, the quantification of these biomarkers[2–6] has demonstrated the ability of optical imaging to detect functional areas. The objective of this work is to supply the methodological tools for the construction of quantitative functional and metabolic brain maps based on the analysis of the cortical reflectance spectra acquired by a hyperspectral camera. The analysis of the ΔC_{HbO_2} and ΔC_{Hb} time courses which are closely linked to the BOLD (blood oxygenation level dependent) contrast used in fMRI studies[7] aim to identify the functional areas of the patient brain. However, these measurements do not monitor metabolism at a cellular level unlike a measurement of the ΔC_{oxCCO}, which has the potential to yield an in vivo indication of cellular energy metabolism.[8]

2. MATERIAL AND METHODS

2.1 Experimental setup

The instrumentation consisted of a hyperspectral camera (*XIMEA MQ022HG-IM-SM5X5-NIR*), a continuous wave white light source (OSRAM Classic 116W 230V Light bulb) and a laptop (processor: Intel Core $i5-7200U$, $2.50GHz \times 4$, ram: $15.3GiB$), see Fig. 1. The camera was used with an Edmund Optics camera lens ($f = 35mm f\# = 1:1:4$) and data were directly acquired by the laptop via an USB link. The sensor provided 16 bits images every 33 ms. The hyperspectral camera captures a complete 3-dimensional hyperspectral cube (hypercube) made up of a matrix of 409×216 pixels and 25 spectral bands ranging from $675nm$ to $1000nm$.

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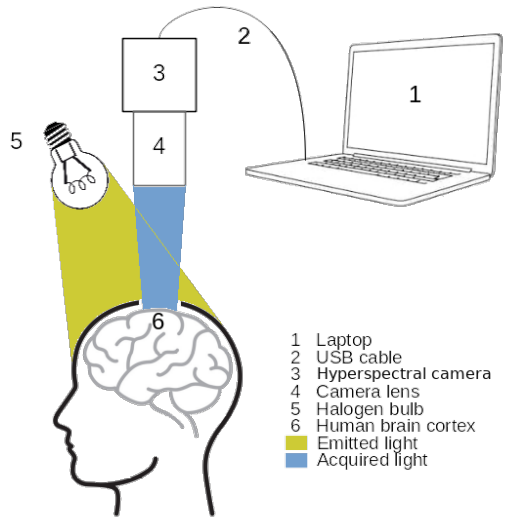

Figure 1. Schematic of the imaging system.

2.2 Patient inclusion and experimental paradigm

The video was acquired during a neurosurgical operation. The patient had his cortex exposed, was awakened and functional tests have been realized. Stimulation of the motor cortex was achieved through successive fist clenching and relaxing. The paradigm consisted of five steps: 20s of rest, 20s of stimulation, 20s of rest, 20s of stimulation and 20s of rest. The neurosurgeon realized electrical brain stimulation to localize patient brain motor areas.

2.3 Data pre-processing

The algorithm developed by *Sdika et al.*[9,10] was implemented to ensure that each camera pixel corresponds to the same cortical area all along data acquisition. The spectral intensities were independently filtered with a low-pass IIR Bessel temporal filter.[1] The slow drift of the collected intensity due to cortical tissue desiccation during the video acquisition[11] was corrected through a linear regression subtracted to the original data.

2.4 Modified Beer Lambert Law

The modified Beer-Lambert law is usually used to convert the acquired intrinsic reflectance spectra into concentration changes. Optical functional brain maps are computed by assessing the oxy and deoxygenated hemoglobin concentration changes.[12–14] Using the near infrared range, the oxidized cytochrome c oxidase concentration changes also can be quantified.[3] This aims to get a metabolic marker of the neuronal activity.[8] The assessment of the concentration changes depends on the determination of the mean optical path length of the detected photons, see Eq. (2). Monte-carlo simulations[15] were processed to estimate the most appropriate mean path length L for each camera pixel.[4,6] The objective of this step is to consider the optical properties inhomogeneities. The modified Beer-Lambert law can be expressed as a linear system,[14] see Eq. (1). In this equation, a 3 chromophore system is represented (deconvolution of ΔC_{HbO_2}, ΔC_{Hb} and ΔC_{oxCCO}).

$$\begin{bmatrix} \Delta A_1 \\ \vdots \\ \Delta A_N \end{bmatrix} = \begin{bmatrix} E_{1,HbO_2} & E_{1,Hb} & E_{1,oxCCO} \\ \vdots & \vdots & \vdots \\ E_{N,HbO_2} & E_{N,Hb} & E_{N,oxCCO} \end{bmatrix} \times \begin{bmatrix} \Delta C_{HbO_2} \\ \Delta C_{Hb} \\ \Delta C_{oxCCO} \end{bmatrix} \tag{1}$$

with

$$E_{i,n} = \int \epsilon_n(\lambda).D_i(\lambda).S(\lambda).L(\lambda).d\lambda. \tag{2}$$

$\Delta A_i(t)$ is the absorbance changes measured at time t by the channel i ($i \in [1;N]$ with $N \in [3;25]$) of the hyperspectral camera:

$$\Delta A_i(t) = \log_{10}\left(\frac{R_i^0}{R_i(t)}\right), \tag{3}$$

where $R_i(t)$ is the reflectance intensity measured at time t by the channel i of the hyperspectral camera, R_i^0 is the reference reflectance intensity measured by the channel i (average of the reflectance intensity over the duration of the first rest step of the experimental paradigm, see Sec. 2.2). ϵ_n is the extinction coefficient of the chromophore n (in $L.Mol^{-1}.cm^{-1}$). The values of ϵ_{Hb} and ϵ_{HbO_2} were taken from the Ref:.[16] The values of ϵ_{oxCCO} were taken from the Ref:.[17] The spectral sensitivity of the detector i of the hyperspectral camera is represented by $D_i(\lambda)$ and $S(\lambda)$ is the normalized intensity spectrum of the light source. $L(\lambda)$ is the wavelength dependent mean optical path length of the photons traveled in tissue. Hemoglobin and cytochrome c oxidaze concentration changes were obtained by matrix inversion once the matrix E has been calculated.

2.5 Hemodynamic and metabolic maps

For each camera pixel, the concentration changes time courses of the chromophore n were used to compute two variables of interest: r_n and $\overline{C_n}$. r_n is the Pearson correlation coefficient computed between the ΔC_n time course and the expected hemodynamic response. The expected hemodynamic response was defined as the convolution of the hemodynamic impulse response function[7] with the window function representing the experimental paradigm, see section 2.2. $\overline{C_n}$ is the ΔC_n value averaged over the patient activity period. Since, the patient had executed several stimulation steps, see section 2.2, the $\overline{C_n}$ values were computed for each stimulation step. The Hb and HbO_2 hemodynamic maps were defined as the cartography of the $\overline{C_{Hb}}$ and $\overline{C_{HbO_2}}$ values on the region of interest (patient exposed cortex). The metabolic map was defined as the cartography of the $\overline{C_{oxCCO}}$ values on the region of interest. A correlation threshold was implemented to display only pixels whose concentration changes time courses were correlated with the expected hemodynamic response.

2.6 Activation map

In our study, an activated cortical area is a cortical area which is associated with the patient hand stimulation, see Sec. 2.2. The ΔC_{Hb} and ΔC_{HbO_2} time courses of an activated cortical area should be highly correlated with the expected hemodynamic response whereas the ones of an non activated cortical area should not. The $\overline{C_{Hb}}$ and $\overline{C_{HbO_2}}$ values of an activated cortical area should be significantly different from those calculated for a non activated cortical areas.

The region of interest of the patient exposed cortex was divided into N cortical areas $Cort$ ($1.8mm$ wide squares). Among these N cortical areas, an area which was defined as non activated by the electrical brain stimulation was selected as a reference area Ref. Successive Welch's T-tests were computed to score the differences between the mean values of the reference area Ref ($< \overline{C_n} >_{Ref}$ and $< r_n >_{Ref}$) and those of the other cortical areas ($< \overline{C_n} >_{Cort}$ and $< r_n >_{Cort}$). The notation $< X >_{Cort}$ corresponds to the mean value of X averaged over the surface of the $Cort$ area.

Each $Cort$ area were tested with the H_0, H_1 and H_2 hypothesis at $1/(N-1)\%$ significance level (Bonferroni correction), see Eqs. (4), (5) and (6).

$$H_0(x) = \begin{cases} \text{True}, & \text{if } x_{Cort} \neq x_{Ref} \\ \text{False}, & \text{otherwise.} \end{cases} \tag{4}$$

$$H_1(x) = \begin{cases} \text{True}, & \text{if } x_{Cort} > x_{Ref} \\ \text{False}, & \text{otherwise.} \end{cases} \tag{5}$$

$$H_2(x) = \begin{cases} \text{True}, & \text{if } x_{Cort} < x_{Ref} \\ \text{False}, & \text{otherwise.} \end{cases} \tag{6}$$

The cortical area $Cort$ was defined as activated if the following condition is respected:

$$Cort = \begin{cases} \text{Activated,} & \text{if } H_0(<\overline{C_{Hb}}>) \,\&\, H_0(<\overline{C_{HbO_2}}>) \,\&\, H_1(<r_{HbO_2}>) \,\&\, H_2(<r_{Hb}>) \\ \text{Non activated,} & \text{otherwise.} \end{cases} \tag{7}$$

3. RESULTS AND DISCUSSION

3.1 Hemodynamic and metabolic maps

In the Fig. 2, the HbO_2, Hb hemodynamic maps and the $oxCCO$ metabolic map are plotted. Only concentration changes time courses which are best correlated with the expected hemodynamic response are considered. The correlation threshold is set to 0.6 for the HbO_2 and Hb maps and is set to 0.4 for the $oxCC0$ map. The highest $\overline{C_{Hb}}$ and $\overline{C_{HbO_2}}$ values are located on the blood vessels surrounding the motor area. The highest r_{HbO_2} and r_{Hb} values are located on the motor area and on the blood vessels surrounding this area. These results correspond to the expected hemodynamic events occurring in the patient cortex following a neuronal activation. The hemodynamic events are strong in the blood vessels and weaker in the grey matter. The highest $\overline{C_{oxCCO}}$ values are located on the grey matter beside the large blood vessels. The highest r_{oxCCO} values are located on the motor area. These results correspond to the expected metabolic events occurring in the patient cortex following a neuronal activation. The cellular metabolism is weak in the blood vessel and stronger in the grey matter.

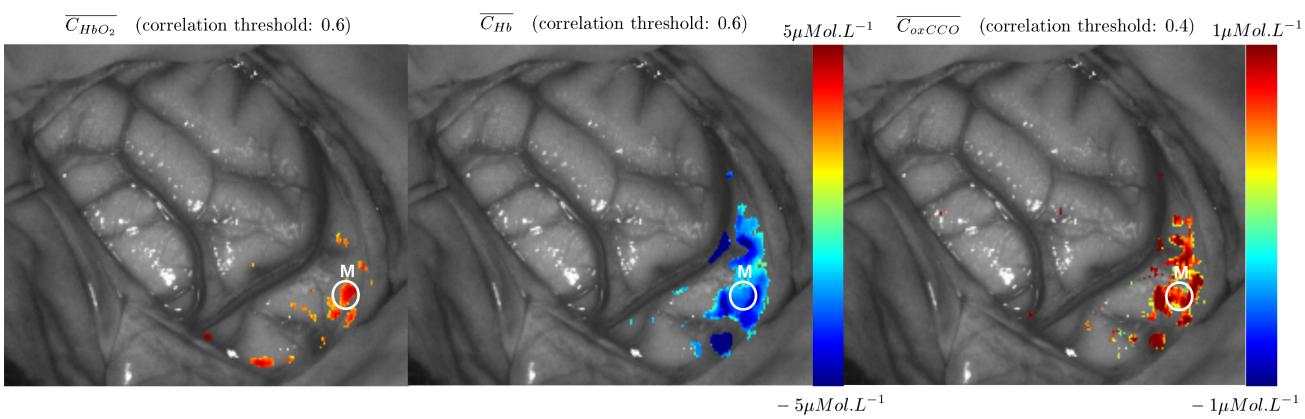

Figure 2. HbO_2, Hb and $oxCCO$ maps plotted on the first spectral image of the hypecube ($675nm$). The HbO_2 and Hb hemodynamic maps were computed with a correlation threshold set to 0.6. The $oxCCO$ metabolic map was computed with a correlation threshold set to 0.4. The markers M designate the motor area identified by electrical brain stimulation. The colorbar represents the scale of variation of the $\overline{C_n}$ values in $\mu Mol/L$.

The ΔC_{HbO_2}, ΔC_{Hb} and ΔC_{oxCCO} time courses averaged over the motor area are represented in the Fig. 3. The ΔC_{HbO_2}, ΔC_{Hb} and ΔC_{oxCCO} time courses are well correlated with the expected hemodynamic response. The r_{HbO_2}, r_{Hb} and r_{oxCCO} values of these time courses are equal to 0.76, 0.86 and 0.84 respectively. The dynamic of the ΔC_{HbO_2} time course is 1.2 times greater than the one of the ΔC_{Hb} time course and 3.7 times greater than the one of the ΔC_{oxCCO} time course. The order of magnitude of these ratios is consistent with the values provided in the literature.[18,19] We can observe that the concentration changes measured during the second activity step (between $t = 67s$ and $= 87s$) are smaller than the ones measured during the first activity step (between $t = 25s$ and $t = 50s$). This may be due to a difference in intensity during the second activity step.

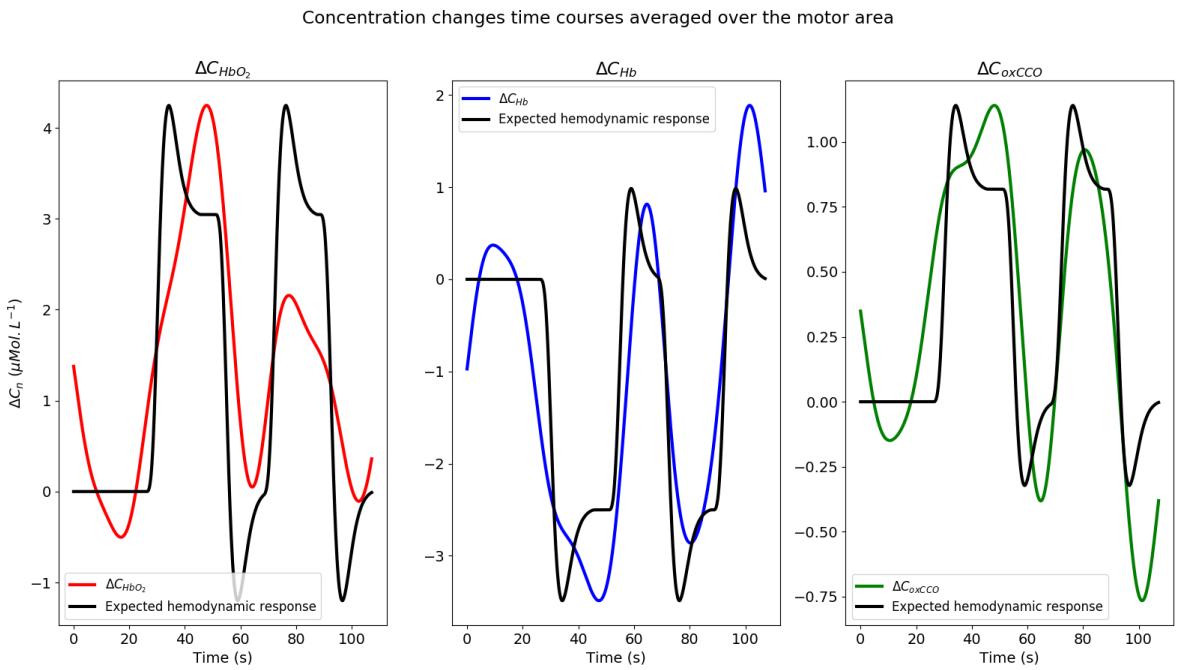

Figure 3. ΔC_{HbO_2}, ΔC_{Hb} and ΔC_{oxCCO} time courses averaged over the motor area, see Fig. 2.

3.2 Activation map

In the Fig. 4, the activated cortical areas are represented in red. The non activated cortical areas are represented in green. The blue square is the reference cortical areas defined as a cortical area that has not been identified as functional by the electrical brain stimulation. The activated cortical areas correspond to the motor area identified by the electrical brain stimulation. Some activated cortical areas are also identified above the motor area (marker M) and at the level of the large blood vessel next to this motor area.

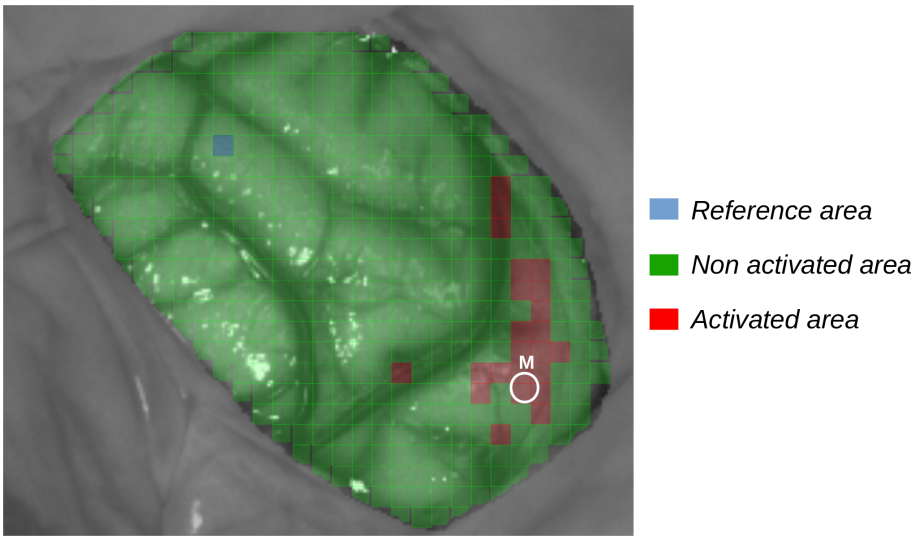

Figure 4. Activation map plotted on the first spectral image of the hypecube ($675nm$). The blue square is the reference area (cortical area that has not been defined as functional by the electrical brain stimulation. The green squares represent non activated areas and the red squares activated areas. The marker M designates the motor area identified by electrical brain stimulation.

This activation model has to be used with the electrical brain stimulation. Indeed, it requires a reference cortical area which is defined as a non functional area by the electrical brain stimulation. The main limits are that the results of the statistical analysis depend of the choice of the reference and the definition of the cortical activation is not obtained for each camera pixel but for a group of pixels. A more robust definition of cortical areas will be explored in future works, notably with the implementation of a *SPM*[20] like analysis. For the moment, the cytochrome c oxidase has not been incorporated in the activation model. Indeed, the theoretical time courses of the ΔC_{oxCCO} following the neuronal activation has not been defined yet.

4. CONCLUSION

A hyperspectral camera and a continuous wave white light illumination seem to be a suitable setup to intraoperatively monitor the functional and metabolic events occurring in a patient exposed cortex following a neuronal activation. These results indicate that our setup could complement the actual intraoperative gold standard for the localization of patient functional areas and could help to better understand the cellular energy metabolism occurring during neurosurgery.

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